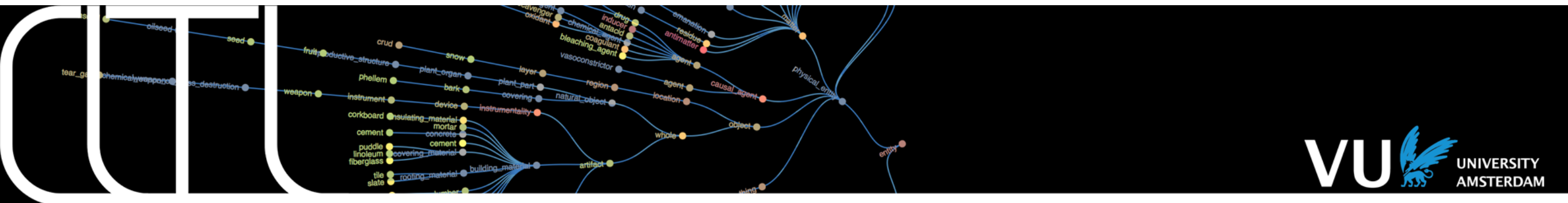


Text Mining 2026



Lecture 4: Subjectivity mining Ilia Markov



Overview

- Part I: What is subjectivity?
- Part II: Subjectivity mining
- Part III: Tools and resources

Part I

What is subjectivity in language?

Part I

What is subjectivity in language?

- Subjectivity in text is a complex phenomenon: all kinds of social and emotional relationships expressed by people when posting information
- bella is the picture of health with **boundless energy** until a few days before **she dies** . this is absolutely and **completely ridiculous** and an **insult** to every **family** whose **mother** has **suffered** through the **horrible pains** of a **death** by **cancer** .

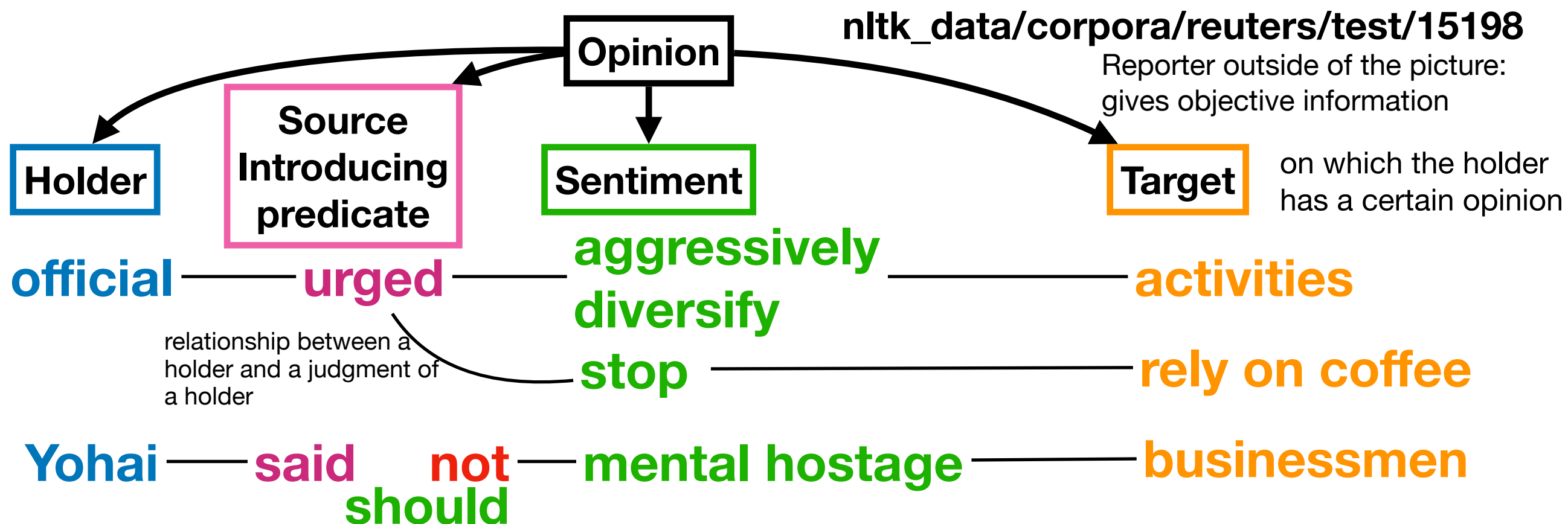
nltk_data/corpora/sentence_polarity/rt-polarity.neg

- **Explicit sentiment:** **boundless energy** (pos), **completely ridiculous, insult, horrible, cancer** (neg)
- **Implicit sentiment:** **dies, suffer, pains, death** (neg?) - in some cultures less
- **Holders:**
 - **Author** (author expresses his opinion, personal message)
 - **Participants:** she, family, mother (emotions not given, inferred from context)
- **Agenda setting:** the author's choice what is mentioned and what is left out (subjective)

Part I

What is subjectivity in language?

- News article, formal style
- A Colombia government trade **official** has **urged** the business community to **aggressively diversify** its **activities** and **stop relying** so heavily on **coffee**. **Samuel Alberto Yohai**, director of the Foreign Trade Institute, INCOMEX, said private **businessmen** **should not** become what he called "**mental hostages**" to coffee, traditionally Colombia's **major** export.



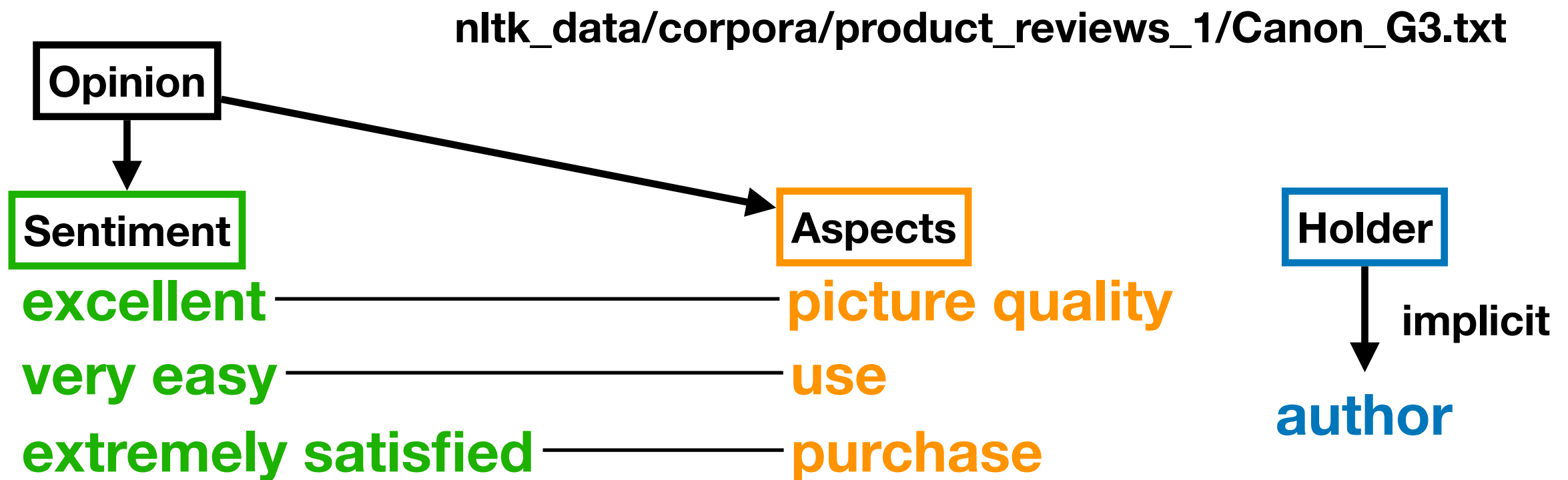
- Opinion is a complex relationship involving a holder and the judgment (sentiment) that the holder has regarding a particular target

Source Introducing Predicates (SIPs)

- Not an open-ended list:
- **Speech-act verbs** (someone is performing a communication act): say, claim, state, express, tell, announce, ensure, guarantee, promise, deny, confirm, reject, accept
- **Cognitive verbs** (mark that someone has a certain mental state): belief, think, consider, feel, hate, like, love
- **Syntactic subject** is an ***agent*** that can be considered as a source of believes (people, organisations), usually who expresses subjective opinion
- **Syntactic object** is a ***proposition***, ***claim*** or ***statement*** (NP, that-clause, VP, e.g., I like taking the train to the university)

Product reviews

- [t]**excellent** **picture quality** / color canon powershot g3[+3]##i recently purchased the canon powershot g3 and am **extremely satisfied** with the **purchase** .
use[+2]##the camera is **very easy** to **use** , in fact on a recent trip this past week i was asked to take a picture of a vacationing elderly group .



Tweets & Irony

- {"contributors": null, "coordinates": null, "text":
"Everything in the **kids section** of
IKEA is **so cute**. **Shame** I'm nearly
19 in 2 months :(“,.....}

nltk_data/corpora/twitter_samples/negative_tweets.json

Figurative language (e.g., metaphors, similes, metonymy), irony, sarcasm are difficult to process, e.g., *It's like going to the dentist*

Emotions as inner states

- Subjective response towards situations (something may trigger or cause an emotion)
- Indirectly inferred from the response: face, voice (tone, pitch, volume), actual content
- What is the vocabulary of emotions?

Ekman et al. (1976)

- 6 Basic emotions in many different cultures (universal status) from facial expressions related to described situations:

- ❖ Anger
- ❖ Disgust
- ❖ Fear
- ❖ Happiness
- ❖ Sadness
- ❖ (Surprise)

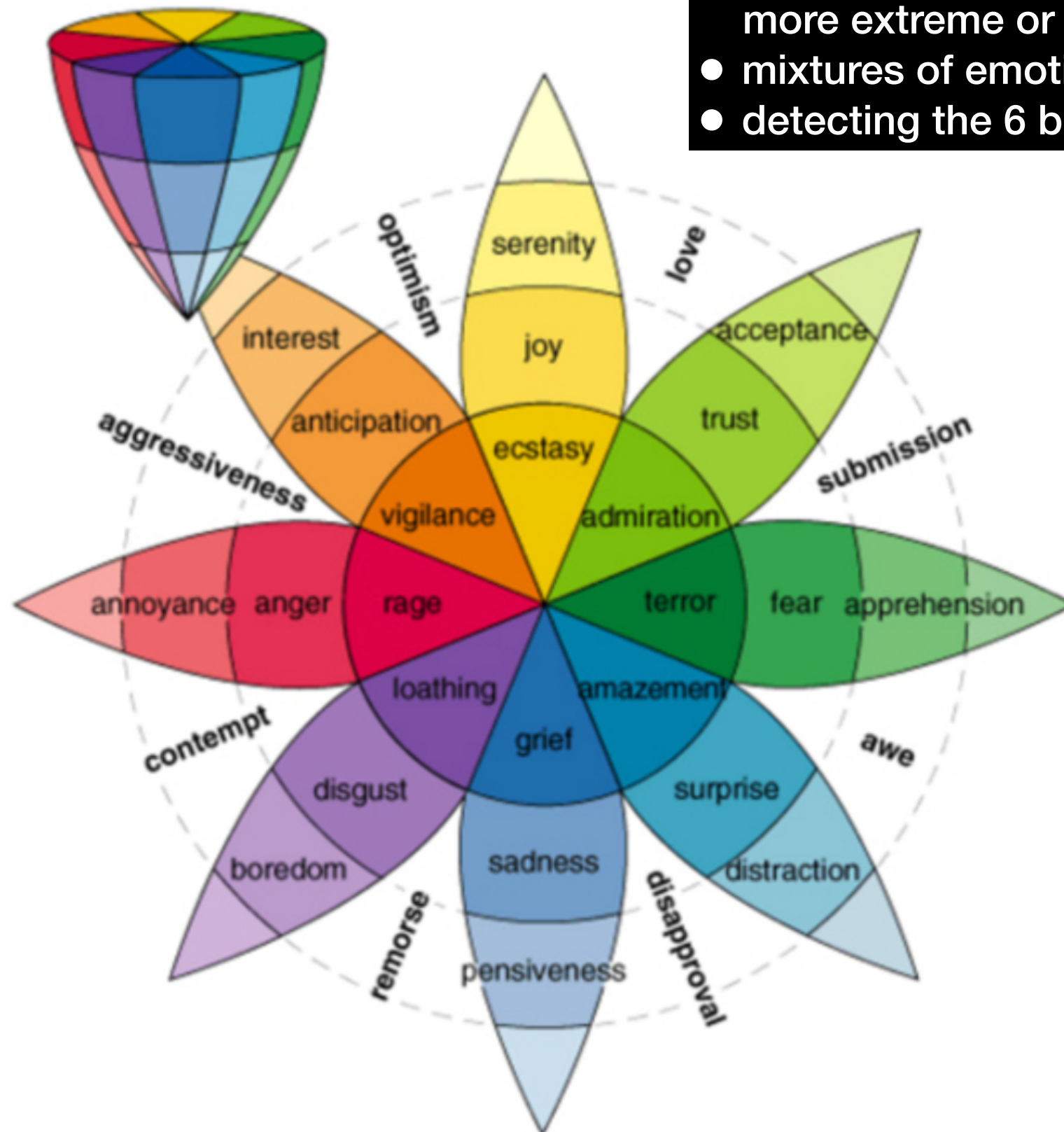
- Extension but not all facially expressed:

- ❖ Amusement
- ❖ Contempt
- ❖ Contentment
- ❖ Embarrassment
- ❖ Excitement
- ❖ Guilt
- ❖ Pride in achievement
- ❖ Relief
- ❖ Satisfaction
- ❖ Sensory pleasure
- ❖ Shame



Plutchik's Wheel of Emotions (1980)

- intensity of basic emotions:
more extreme or more intense emotions
- mixtures of emotions
- detecting the 6 basic emotions is not trivial



<http://wndomains.fbk.eu/wnaffect.html>

Emotions and other mental states (for English)

A-Labels and corresponding example synsets

A-Labels	Examples
EMOTION	<i>noun anger#1, verb fear#1</i>
MOOD	<i>noun animosity#1, adjective amiable#1</i>
TRAIT	<i>noun aggressiveness#1, adjective competitive#1</i>
COGNITIVE STATE	<i>noun confusion#2, adjective dazed#2</i>
PHYSICAL STATE	<i>noun illness#1, adjective all in#1</i>
HEDONIC SIGNAL	<i>noun hurt#3, noun suffering#4</i>
EMOTION-ELICITING SITUATION	<i>noun awkwardness#3, adjective out of danger#1</i>
EMOTIONAL RESPONSE	<i>noun cold sweat#1, verb tremble#2</i>
BEHAVIOUR	<i>noun offense#1, adjective inhibited#1</i>
ATTITUDE	<i>noun intolerance#1, noun defensive#1</i>
SENSATION	<i>noun coldness#1, verb feel#3</i>

Terminology

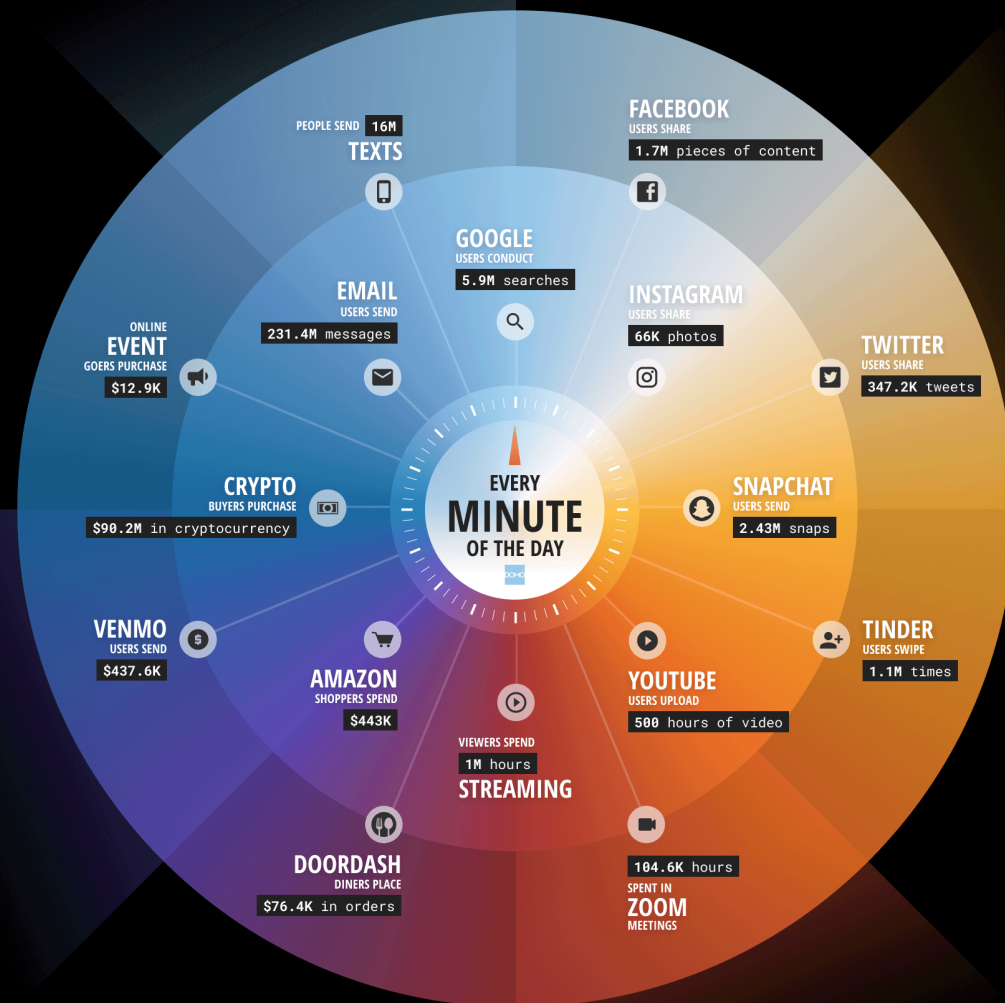
Subjectivity or Attitude	broad term that covers all forms of opinion mining, sentiment analysis, stance but leaves open the lexical or structural realisation (what expressions go with it)
Sentiment or Polarity	explicit expression of being positive, negative or neutral about something/ someone : good , bad
Opinion	a lexically or syntactically realised sentiment relation of a holder to a target: I like him, my hero failed
Stance	opinion in a debate: I support the anti-vaccination movement
Aspects-facets-features-properties	opinion on an aspect of something such as products: this iPhone has an incredible good battery
Argumentation	providing explicit arguments for stances: I like this iPhone because it has a very powerful battery
Emotion	response to a situation that is deemed important: anger, joy, sadness
Attribution	relation between a source , a cue (not differentiated) and some content , e.g. quotes, direct/indirect citations: The BMW spokesman said: “....” Used for connecting a sentiment/opinion to a source (alternative term to holder in case of opinion + authorship)
Other terms	perspective, inner state, mental state, ...



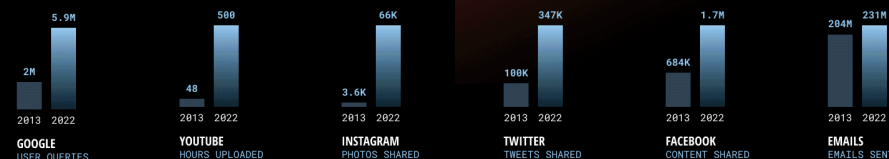
DATA NEVER SLEEPS 10.0

Over the last ten years, digital engagement through social media, streaming content, online purchasing, peer-to-peer payments and other activities has increased hundreds and even thousands of percentage points. While the world has faced a pandemic, economic ups and downs, and global unrest, there has been one constant in society:

our increasing use of new digital tools to support our personal and business needs, from connecting and communicating to conducting transactions and business. In this 10th annual "Data Never Sleeps" infographic, we share a glimpse at just how much data the internet produces each minute from some of this activity, marveling at the volume and variety of information that has been generated.



DATA NEVER SLEEPS 1.0 VS. 10.0



GLOBAL INTERNET POPULATION GROWTH



As of April 2022, the internet reaches 63% of the world's population, representing roughly 5 billion people. Of this total, 4.65 billion - over 93 percent - were social media users. According to Statista, the total amount of data predicted to be created, captured, copied and consumed globally in 2022 is 97 zettabytes, a number projected to grow to 181 zettabytes by 2025.

To succeed in an increasingly digital world where the volume of data created keeps accelerating, businesses need the right tools to put that data to work right where work gets done. Domo gives you the power to rapidly unlock value from all your data, regardless of where it lives, and drive actions across your organization that will improve business outcomes. Every click, swipe, share, or like tells a story, and Domo helps you do something powerful with it.

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SOURCES

Global Media Insight, Oberlo, Hootsuite, Earthweb, Matthew Woodward.co.uk, Web Tribunal, Deadline.com, Local IQ, Business of Apps, Query Sprout, Young and the Invested, Dating Zest, Ibis World, DoorDash, TechCrunch, Statista, Data Never Sleeps 1.0



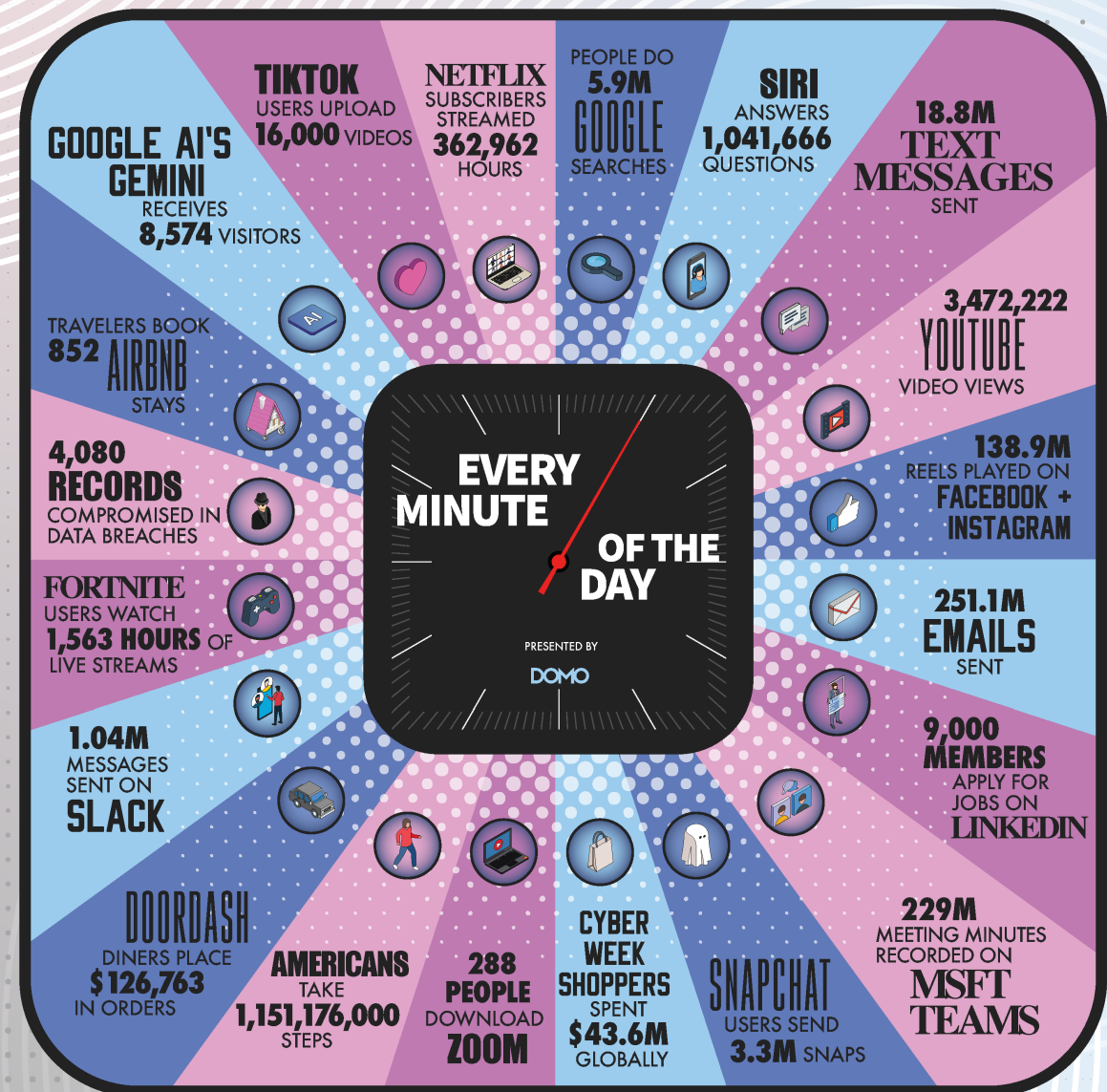
DATA NEVER SLEEPS



Every minute of every day, the world generates a dizzying amount of data, and how we interact with it is constantly changing. AI tools are now answering millions of questions in real time and are transforming how we work, shop, and connect. Digital platforms are seeing explosive usage, with billions of emails, texts, and reels shared every day. Entertainment continues to drive engagement across streaming, gaming, and social media while e-commerce is setting new benchmarks as digital habits evolve and expand at an unprecedented pace.

In Domo's 12th edition of Data Never Sleeps, we capture a snapshot of this world powered by the rapid rise of data, AI, and digital activity, shaping every moment of modern life.

12



The world's internet population continues to grow significantly year-over-year. As of late 2024, 5.52 billion people—approximately 67.5% of the global population—are online.

According to industry analysts, the total amount of data created, captured, copied, and consumed globally is expected to reach 149 zettabytes by the end of 2024, with projections surpassing 394 zettabytes by 2028.

Global Internet Population Growth (IN BILLIONS)



As the volume and complexity of data accelerates, business success increasingly depends on the ability to turn information into insights. Domo helps you harness the power of data and AI so you can adapt as quickly as the world changes and make data-driven decisions that set you apart. Let Domo help you make sense of all the clicks, swipes, and streams so you can see the big picture shaped by every small decision.

Learn more at domo.com

SOURCES: EARTHWEB, DUSTIN STOUT, DEMANDSAGE, HOOTSUITE, BUSINESSOFAPPS, DOORDASH, SOCIALPILOT, X | TWITTER.COM, GITLUX, INVIGATE, THINKIMPACT, SIFMA.ORG, STATISTA, PR NEWS-WIRE, NETSCOUT



We do not live in an information society

...

We live in a **communication** society

...

Any kind of subjectivity mining relates to a
communication society



Part II

What is Subjectivity Mining?

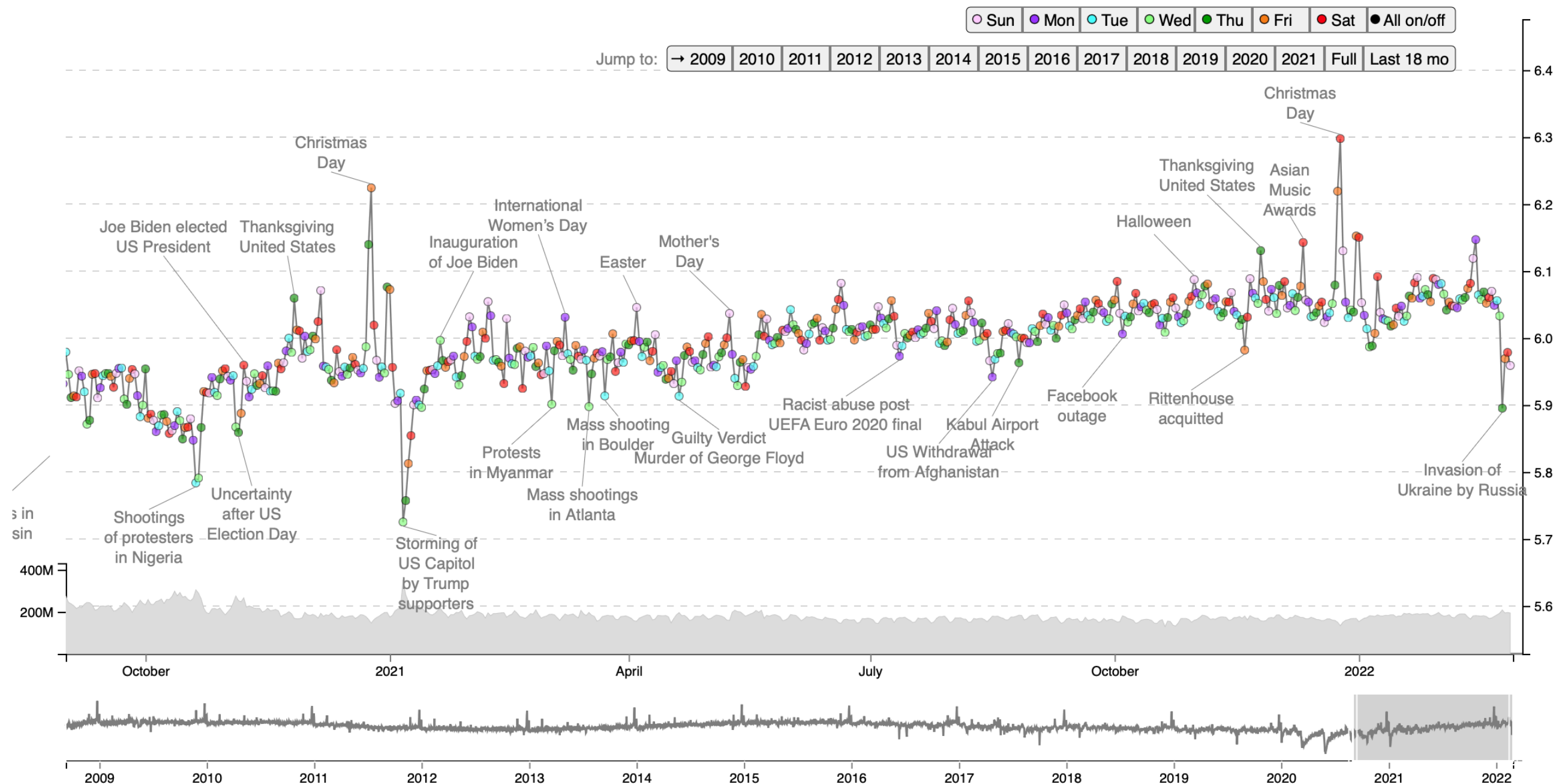
- Software for automatically extracting **opinions, emotions and sentiment** in text.
- It allows us to track attitudes and feelings on the web. People write **blogs, comments, reviews and tweets** about all sorts of different topics.
- We can track products, brands and people, and determine whether they are **positively or negatively** perceived on the web (reputation management).

hedonometer.org

Correlation between events and the sentiment reflected in the majority of the tweets:
below negatively, on top positively classified.

Average Happiness for Twitter

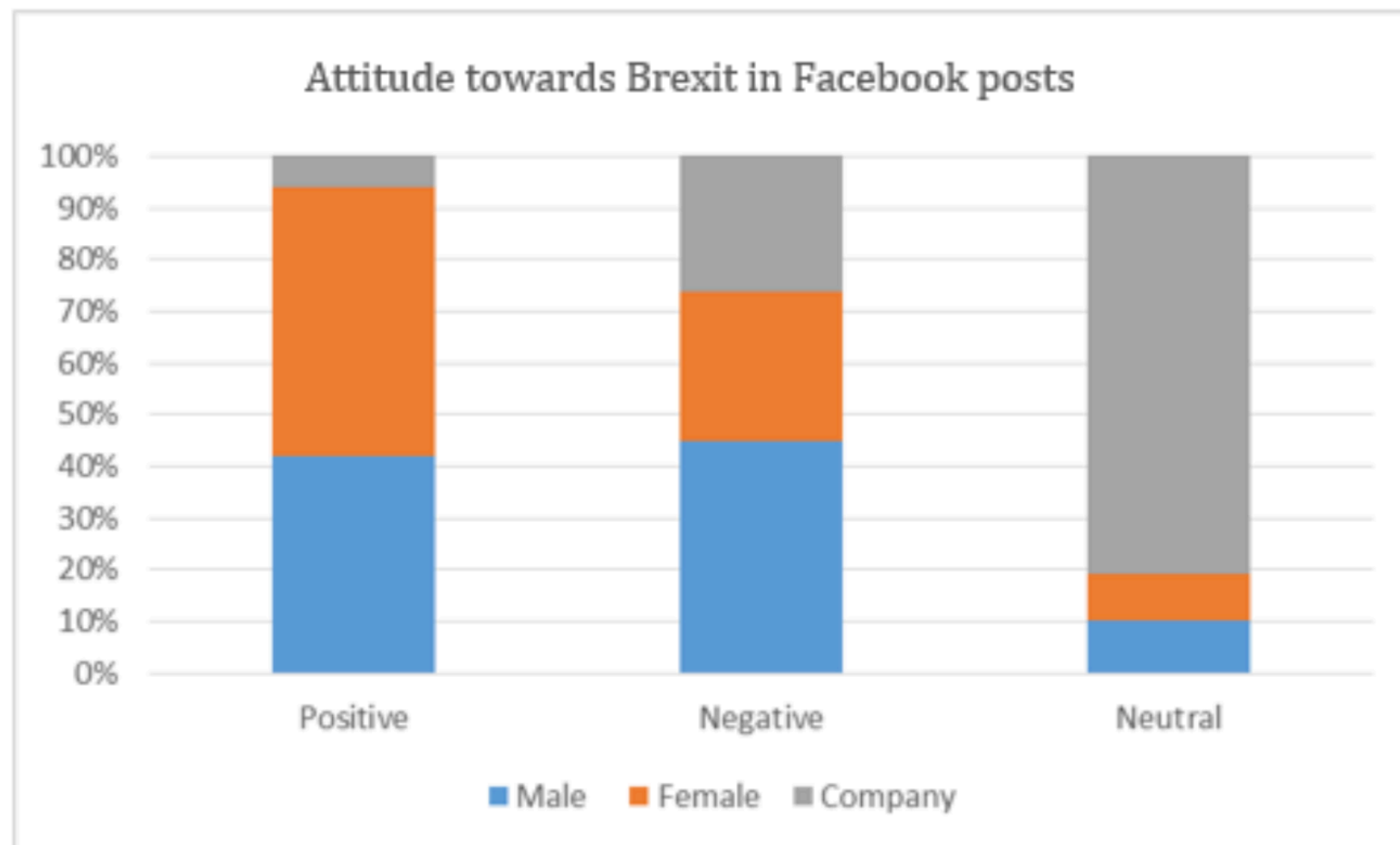
All Tweets in English.



pro/against Brexit (2016)

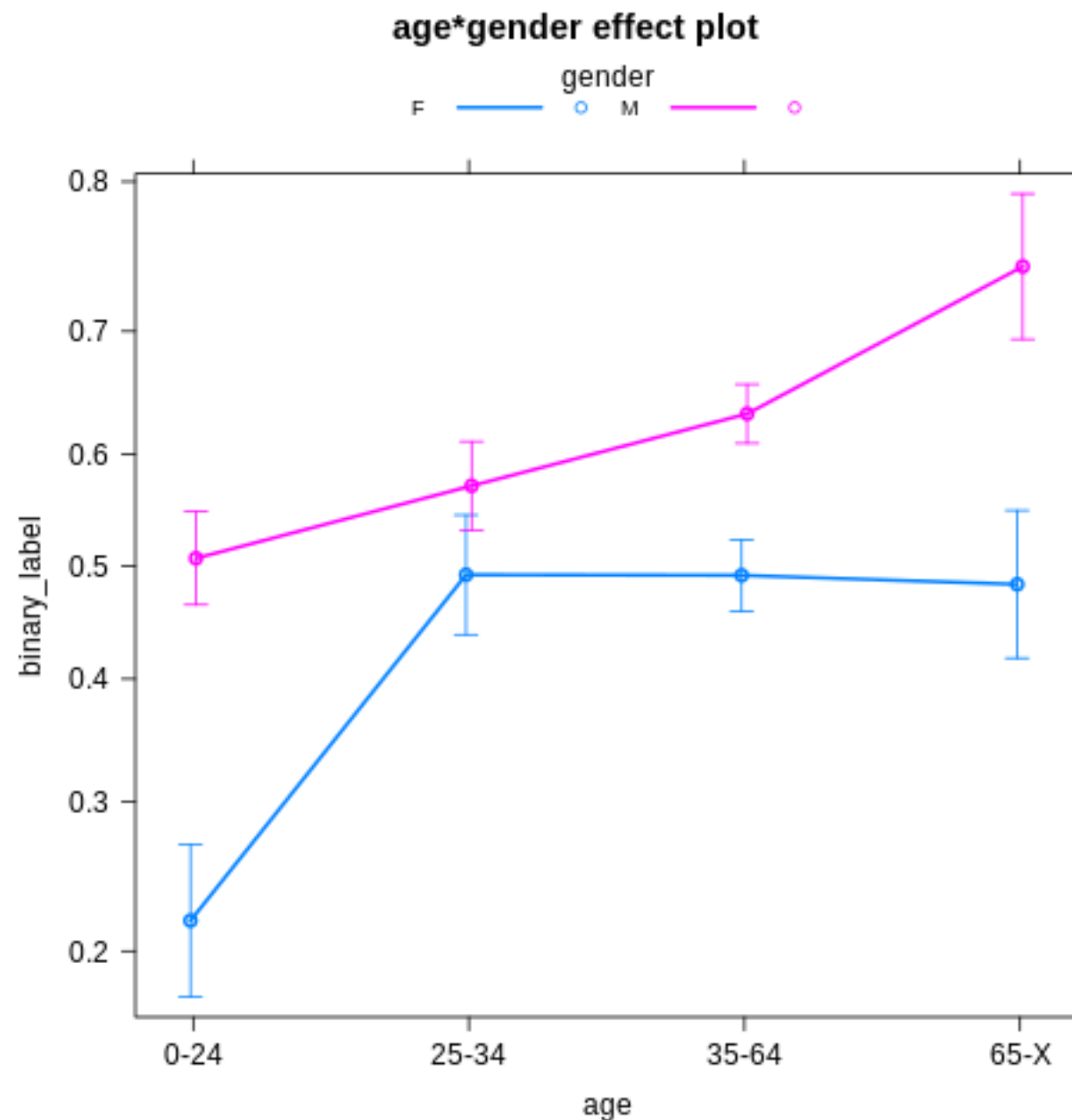
Societal analysis: stance w/ respect to Brexit correlated with gender

Interestingly, females were most likely to post **for** Brexit (68%) followed by men (55%), while companies posted mainly neutral content (71%). 36% of the posts made by men were actually **against** Brexit while only 23% of women voiced a negative opinion.



Demographics Analysis of Hateful Content Creators

Effect plot for age, gender w.r.t. posting hateful content in Dutch
(Hilte et al., 2023)



Levels of Analysis

- **Collection of documents** (corpus): emotions in novels before or after WW-II, 9/11, COVID-19, Russian invasion in Ukraine (whether what happens in the world has an impact on the emotion/sentiments/opinions that people express)
- **Document:** one sentiment per document
- **Sentence:** single document may hold several opinions on several topics by several opinion holders expressed in sentences
- **Phrase:** within a sentence we find opinions on entities, topics or aspects (e.g., what aspects of an hotel are being positively or negatively perceived)

=> Different techniques and text representations

Sentiment analysis steps

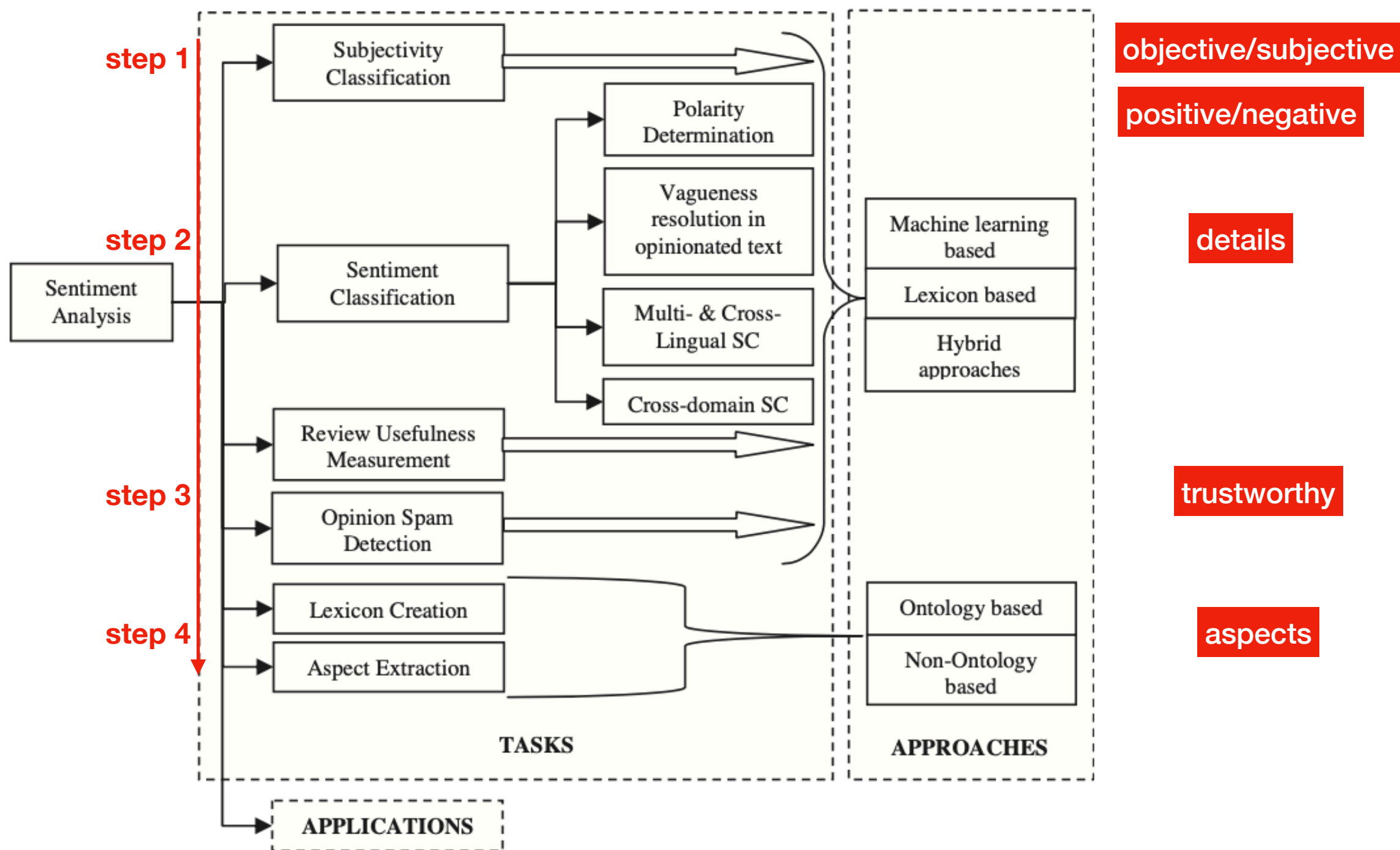


Fig. 1. Organization of the review.

Methods

- Opinion extractions as a set of different classification problems:
 1. Is the sentence subjective (yes or no)?
 2. What is the opinion expression?
 3. Who is the holder of the opinion?
 4. What is the target of the opinion?
 5. Is it negative or positive?
- Joint inferencing (all tasks in one system) or separate classification tasks
- Rule-based, Supervised machine learning (combined with unsupervised learning)

Features for sentiment mining

- **Features:**

- (Bags of) words or n-grams
- Part-of-speech (e.g., adjectives and adjective-adverb combinations)
- Lists of opinion words
- Valence intensifiers and shifters (negation); modal verbs; ...
- Syntactic dependencies (for opinions): subject, object of Source Introducing Predicates

- **Feature selection based on**

- frequency, information value (TF*IDF)
- term position: e.g., title, first and last sentence(s)

- **One hot-encoding of features in texts:**

- maintains details about wordings & expressions

- **Averaged word-embeddings**

- more semantic and associative (stronger generalization), potentially higher recall but less precision

- **Contextual-embeddings (transformer encoders such as BERT):**

- best of both worlds: semantic representation and compositional relations through attention (better recall and precision)

Representing data

- *The rooms are not clean and do not have their own bathroom*
- PoS: Det N V NEG Adj CO V NEG V Det A N
- IOB opinions: B-target I-target O B-opinion I-opinion O O B-opinion I-opinion B-target I-target I-target
- Sequence of sentiment words: O O O NEG POS O O NEG POS POS O
- one-hot-BoW-encoding: $[0, 1, 0, 0, 1, 0, 1, \dots \text{etc} \dots]$
- embedding-token-encoding: $[[200d]_1, [200d]_2, \dots, [200d]_{12}]$
- mean-sentence-embedding: $[.354, .03344, .4434, .654, .004]_{768}$

Machine Learning methods

Table 6
Distribution of articles based on intelligent techniques applied.

Applied techniques	#Articles	Articles' references
SVM	55	[21,26,29,33,44–46,50,51,53,54,57,58,66,67,73,76,77,86,88,90,91,94,95,97,101,108,109,111,114,116,118,125,131,148,157,160,163,165,167,169,172,176,177,183,195,197,200,209,210,212,214,225,228,240]
Dictionary based approaches (DBA)	41	[13,18,23–25,35,36,47,55,64,67–69,85,96,110,112,117,126,127,158,170,171,175,183,193–196,202,203,206,207,209,210,213,216,218,220,229,241]
NB	28	[33,46,50,54,56,73,80,86,90,94,98,101,111,114,116,118,121,124,125,131,148,156,163,167,197,209,217,228]
NN	11	[48,50–53,57,76,101,116,213,226]
DT	9	[53,66,73,76,86,94,116,118,209]
Maximum entropy	8	[33,46,54,60,63,66,148,156,174]
Logistic regression	9	[53,77,88,99,116,118,163,197,220]
Linear regression	8	[8,18,70,75,222–224,231]
Ontology	8	[30,41,62,98,182,194–196]
LDA	8	[61,92,107,185,189,191,182,240]
Random forest	4	[77,81,228,210,228]
SVR	5	[118,123,130,155,227]
CRF and rCRP	5	[37,88,93,186,190]
Boosting	4	[12,118,179,197]
SVM-SMO	4	[76,97,118,169]
Fuzzy logic	3	[23,41,62,213]
Rule miner	4	[37,100,112,217]
EM	3	[56,59,155]
K-medoids	1	[52]
RBF NN	1	[130]

Table 5
Sentiment classification accuracy reported on common datasets.

S#	Dataset	Articles	Obtained result
1	Pang and Lee [167]	[156]	92.70% accuracy
2		[112]	90.45% F ₁
3		[169]	90.2% accuracy
4		[35]	89.6% accuracy
5		[54]	87.70% accuracy
6		[46]	87.4% accuracy
7		[50]	86.5% accuracy
8		[26]	85.35% accuracy
9		[162]	81% F ₁
10		[124]	79% accuracy & 86% F ₁
11		[61]	76.6% accuracy
12		[69]	76.37% accuracy
13		[48]	75% precision
14		[98]	79% precision
15	Pang et al. [33]	[109]	Approx. 90% accuracy
16		[165]	88.5% accuracy
17		[172]	87% accuracy
18		[33]	82.9% accuracy
19		[156]	78.08% accuracy
20		[180]	75% accuracy
21		[48]	60% precision
22		[195]	86.04%
23	Blitzer et al. [149]	[45]	84.15% accuracy
24		[99]	80.9% (Avg.) accuracy
25		[54]	85.15% (Avg.) Max. 88.65% accuracy on Kitchen reviews
28		[165]	88.7% accuracy
29		[61]	71.92% accuracy

SVM the most used (2015!)
Results 72-92% in-domain accuracy

Ravi & Ravi 2015

Document level sentiment classification with and without preprocessing

F-score

- Ex. (2) *The rooms are not clean and do not have their own bathroom*

- Rule and lexicon-based: negation flips vote

78.3

- Machine Learning Naive Bayes, 1,171 hotel reviews

- Bag-of-WORDS

82.3

- Bag-of-WORDS + LEMMA

83.6

- Bag-of-WORDS + sentiment lexicon and rules:

84.7

- the rooms are not clean+**negator_tag** and do not have their own bathroom+**negator_tag**

Performance up to 92% (combining features, optimising parameters)

Do we agree on opinions?

(annotations at CLTL)

- **Hotel reviews**
 - Document level: 3 classes (positive, negative, neutral),
Cohen-kappa: 0.87 (high, takes chance into account)
- **Black Pete debate** (more complex)
 - Tweet level: 4 classes / 3 annotators,
Cohen-kappa: 0.65 (a1-a2); 0.57 (a1-a3); 0.55 (a2-a3)
- **News annotations** (complex relationships: journalists quote people to write about particular positions or opinion)
 - opinion expressions, Cohen-kappa 0.70

Personal perspective of annotators plays a role

1-5

-5, 0, +5

Review text	User rating	Ann1	Ann2
The hotel is fantastic , but the area around the hotel is very very noisy .	4	-2	3
Good Hotel, in Down Town. Really for business. For holidays I had choose something else .	4	-2	-2

Opinion Annotation

Washington (AFP) - Obama said Thursday that he had included openly gay athletes in the US Olympic delegation to show the **United States** would **not abide discrimination** in sport or anywhere else.

His comments, in an interview with NBC on the eve of the Sochi Winter Olympics, came after a **senior Russian official** activists should **not promote gay rights** during the Olympics following the passage of controversial anti-gay legislation in Russia.

Obama **picked** several **openly gay former athletes** in the US delegation to the opening of the games and pointedly did not dispatch a cabinet-level official or a member of his family to Sochi

32

20

4

Context dependence

Difficult to generalize sentiment from one area to another area

- A **cold** person
- A **cold** soda
- A **cold** shower
- Low prices or ceilings
- Fast food or cars

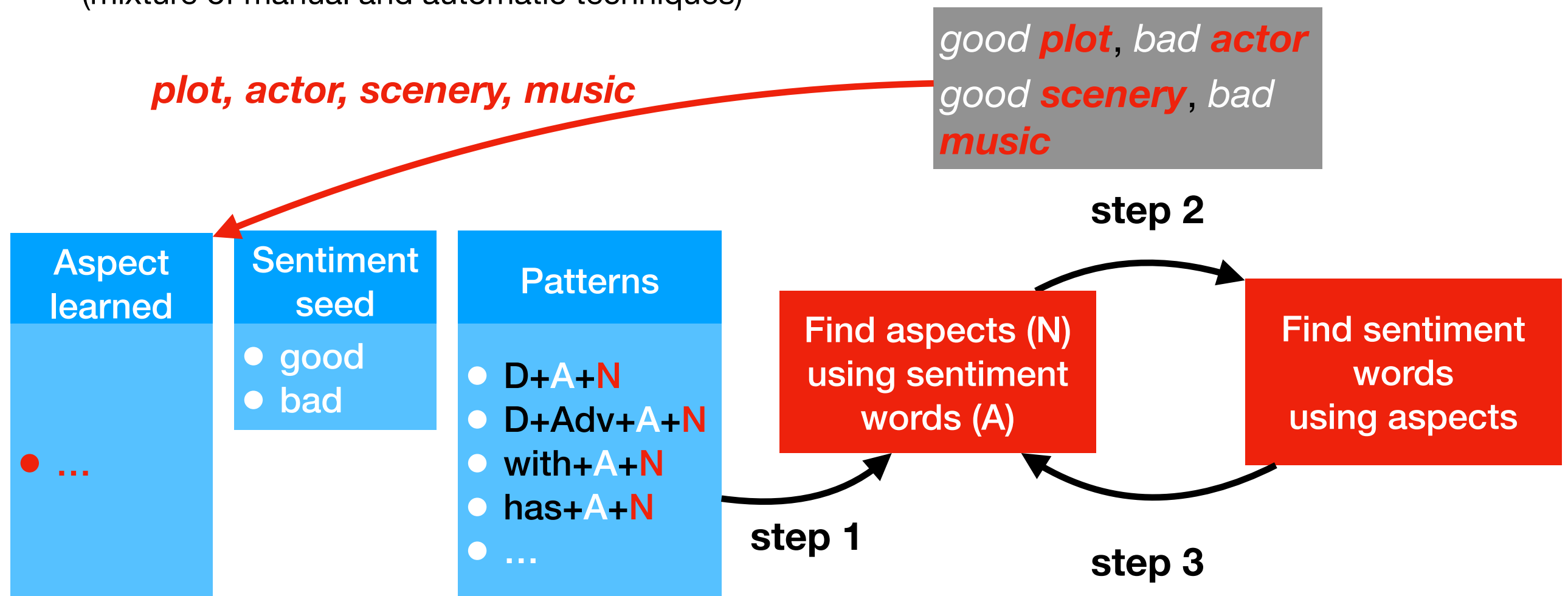
How to learn the sentiment value from specific datasets?

Weak supervision or double propagation

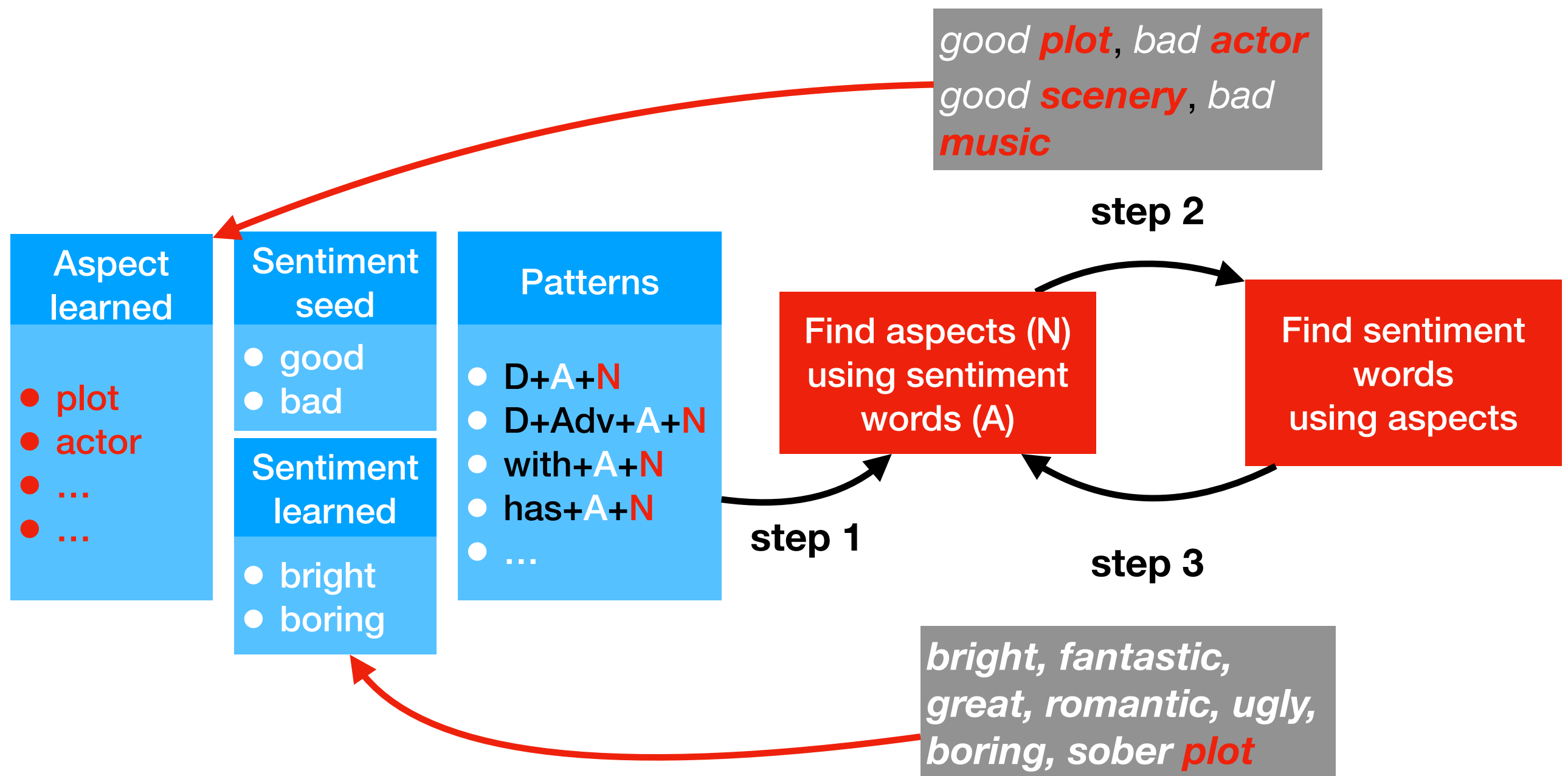
Double propagation:

Learning sentiments & aspects from data

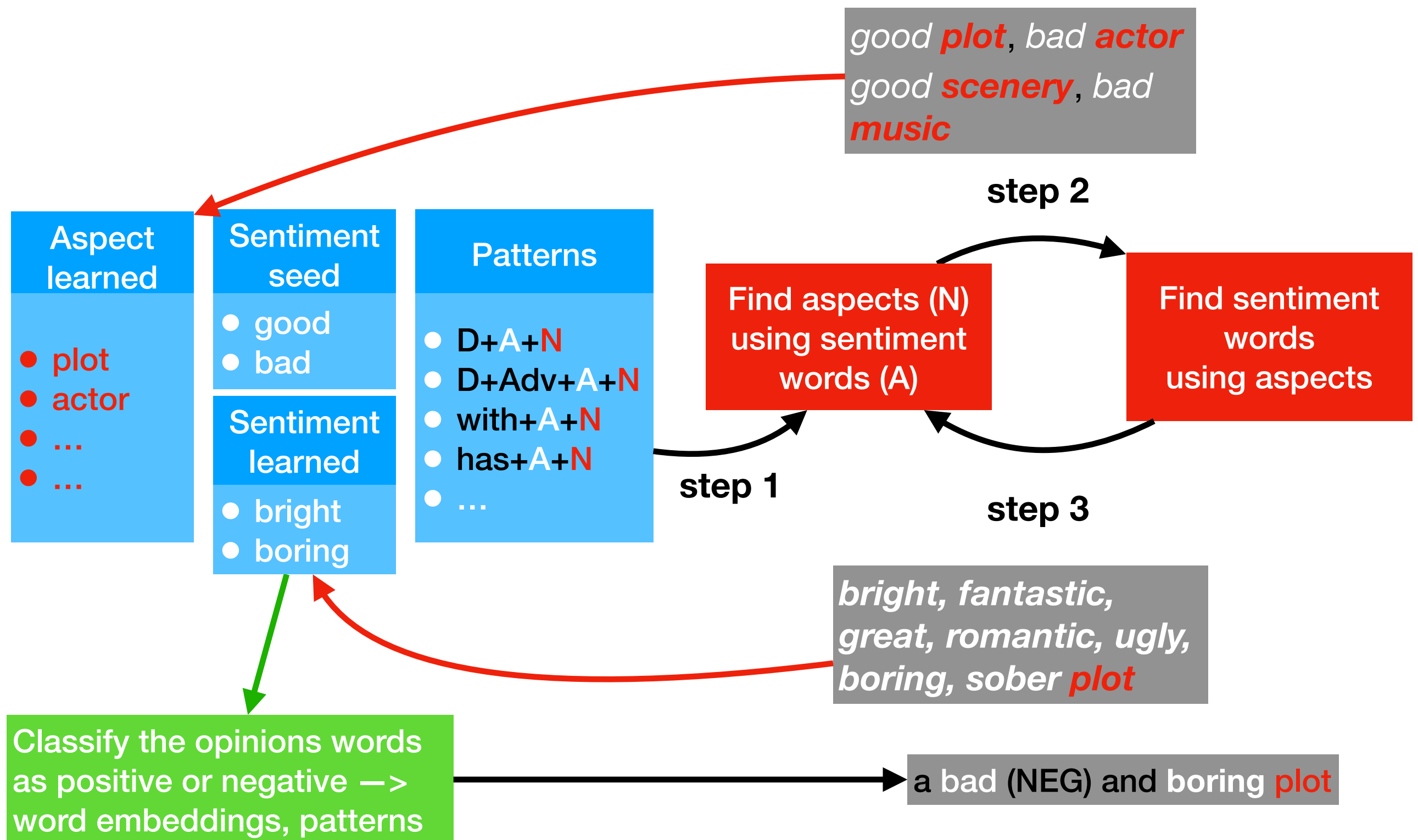
(mixture of manual and automatic techniques)



Double propagation: Learning sentiments & aspects from data



Double propagation: Learning sentiments & aspects from data



Aspect-Based Sentiment Analysis

Aim: learn particular aspects people have an opinion about

Example of XML data:

- SemEval tasks:

- <https://alt.qcri.org/semeval2014/task4/>
- <https://alt.qcri.org/semeval2016/task5/#>

```
<Review rid="1004293">
```

```
<sentences>
```

```
<sentence id="1004293:0">
```

```
<text>Judging from previous posts this used to be a good place, but not any longer.</text>
```

```
<Opinions>
```

```
<Opinion target="place" category="RESTAURANT#GENERAL" polarity="negative"
  from="51" to="56"/>
```

```
</Opinions>
```

```
</sentence>
```

```
<sentence id="1004293:3">
```

```
<text>The food was lousy - too sweet or too salty and the portions tiny.</text>
```

```
<Opinions>
```

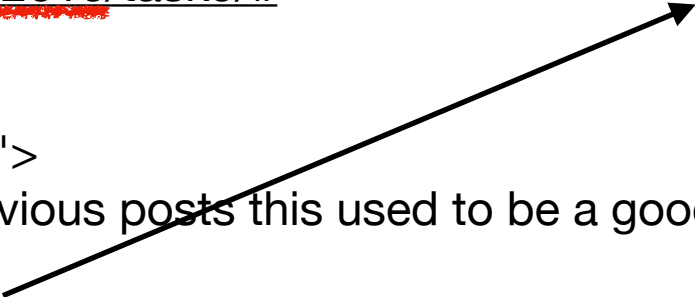
```
<Opinion target="food" category="FOOD#QUALITY" polarity="negative"
  from="4" to="8"/>
```

```
<Opinion target="portions" category="FOOD#STYLE_OPTIONS" polarity="negative"
  from="52" to="60"/>
```

```
</Opinions>
```

```
</sentence>
```

Target of the judgement (aspect)



Train a system to classify each token either being an aspect or not (token level, e.g., BIO format), categorize which kind of aspect it is (restaurant, food). Sentiment task is another classification on top of it (sentence level or more complex: opinion extraction).

Top 15 ranked aspect terms for restaurants	Top 15 ranked aspect terms for laptops	Top 15 ranked aspect terms for hotels
1- food	1- battery life	1- hotel
2- service	2- keyboard	2- room
3- staff	3- screen	3- staff
4- bar	4- feature	4- service
5- drink	5- price	5- food
6- table	6- machine	6- view
7- menu	7- toshiba laptop	7- stay
8- dish	8- windows	8- breakfast
9- atmosphere	9- performance	9- pool
10- pizza	10- use	10- floor
11- meal	11- battery	11- area
12- bartender	12- program	12- location
13- price	13- speaker	13- bed
14- server	14- key	14- beach
15- dinner	15- hard drive	15- bar

Aspect extracted by token classification systems: different aspects depending on the type; restaurants and hotels share different aspects. Take into account for different types of products.

Tools & Resources

VADER lexicon

Valence Aware Dictionary and sEntiment Reasoner

TOKEN, MEAN-SENTIMENT-RATING, STANDARD DEVIATION, and RAW-HUMAN-SENTIMENT-RATINGS [10 raters -4, 0, +4]

```
arrogance    -2.4    0.66332 [-3, -2, -2, -3, -1, -2, -3, -3, -2, -3]
arrogances   -1.9    0.53852 [-1, -2, -3, -2, -2, -2, -2, -1, -2, -2]
arrogant     -2.2    0.6    [-2, -2, -2, -3, -2, -3, -3, -2, -1, -2]
arrogantly   -1.8    1.4    [-3, -2, 2, -3, -1, -3, -2, -2, -2, -2]
ashamed      -2.1    1.3    [-3, -3, -3, -2, -2, 1, -2, -4, -1, -2]
ashamedly    -1.7    0.64031 [-2, -2, -2, -1, -2, -3, -1, -2, -1, -1]
ass          -2.5    1.43178 [-4, -1, -2, -1, -3, 0, -2, -4, -4, -4]
```

- 😊 grinning face
- 😄 beaming face with smiling eyes
- 😂 face with tears of joy
- 😆 rolling on the floor laughing
- 😁 grinning face with big eyes
- 😊 grinning face with smiling eyes
- 😓 grinning face with sweat
- 😏 grinning squinting face
- 😉 winking face

Tools & Resources

NRC emotion lexicon

<https://saifmohammad.com/WebPages/NRC-Emotion-Lexicon.htm>

● beating	anger	1
● beating	anticipation	0
● beating	disgust	0
● beating	fear	1
● beating	joy	0
● beating	negative	1
● beating	positive	0
● beating	sadness	1
● beating	surprise	0
● beating	trust	0

● betrayal	anger	1
● betrayal	anticipation	0
● betrayal	disgust	1
● betrayal	fear	0
● betrayal	joy	0
● betrayal	negative	1
● betrayal	positive	0
● betrayal	sadness	1
● betrayal	surprise	0
● betrayal	trust	0

Tools & Resources

- Sentiment for Tweets: VADER (for tweets); <https://github.com/cjhutto/vaderSentiment>
- NRC (emotions expressed by words): <https://saifmohammad.com/WebPages/NRC-Emotion-Lexicon.htm>
- Linguistic Inquiry and Word Count, Psychological features: <http://liwc.wpengine.com>
- WordNet Affect: <http://wndomains.fbk.eu/wnaffect.html>
- MPQA, OpinionFinder: subjective sentences, source (holder) of the subjectivity and words that are included in phrases expressing positive or negative sentiments: <http://mpqa.cs.pitt.edu>
- Bing Liu's opinion lexicon: <http://www.cs.uic.edu/~liub/FBS/sentiment-analysis.html>
- Basic sentiment tokenizer plus some tools, by Christopher Potts: <http://sentiment.christopherpotts.net>
- Harvard General Inquirer: <http://www.wjh.harvard.edu/~inquirer/>

(Some) Tools

- SentiStrength (sentistrength.wlv.ac.uk)
- TheySay (apidemo.theysay.io)
- Sentic (sentic.net/demos)
- Sentdex (sentdex.com)
- Lexalytics (lexalytics.com)
- Sentilo (wit.istc.cnr.it/stlab-tools/sentilo)
- nlp.stanford.edu/sentiment

Overview

sentiment datasets (2015)

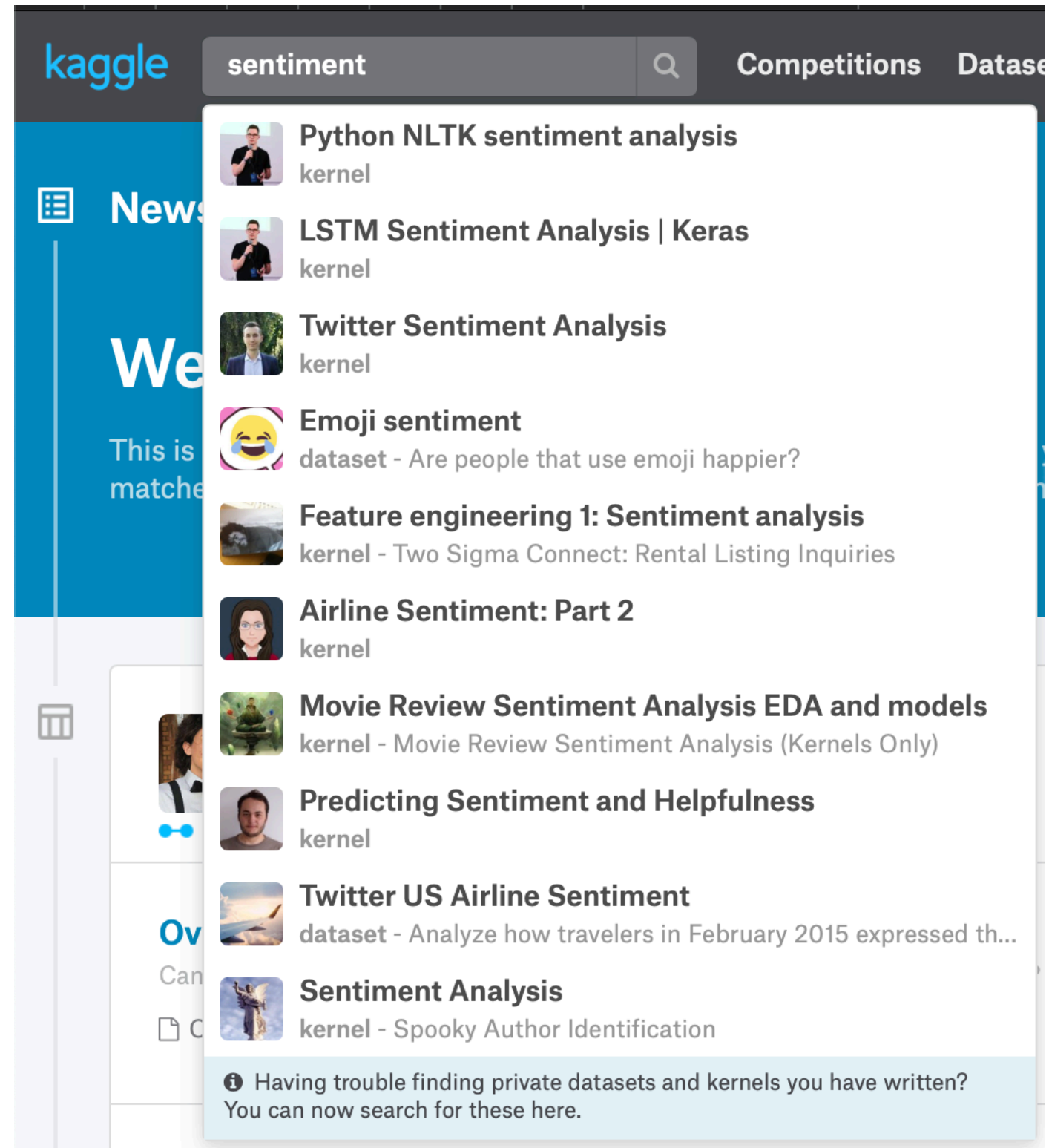
Table 9

List of publicly available datasets.

S#	Data set	Type	Lang.	Web resource	Details
1	Stanford large movie data set	Movie Reviews	English	http://ai.stanford.edu/~amaas/data/sentiment/	Movie Reviews
2	COAE2008	Product Reviews	Chinese	http://ir-china.org.cn/coae2008.html	2739 documents for movie, education, finance, economics, house, computer, mobile phones, etc. 1525 +ve, 1214 -ve
3	Boacar	Car Reviews	Chinese	http://www.riche.com.cn/boacar/	11 type of car TradeMarks and total review 1000 words, having 578 POS, 428 -ve reviews
4	[187]	Reviews, forums	English	http://sifaka.cs.uiuc.edu/~wang296/Data/	Accessed: 27 August, 2014
5	[188]	Reviews	English	http://uilab.kaist.ac.kr/research/WSDM11	Aspect oriented dataset. Accessed: 18 December, 2014
6	Movie-v2.0	Movie Reviews	English	http://www.cs.cornell.edu/people/pabo/movie-review-data/	Data size: 2000 Positive: 1000 Negative: 1000
7	Multi-domain	Multi-domain	English	http://www.cs.jhu.edu/~mdreze/datasets/sentiment	
8	SkyDrive de Hermit Dave	Spanish Word Lists	Spanish	https://skydrive.live.com/?cid=3732e80b128d016f&id=3732E80B128D016F%213584	
9	TripAdvisor	Reviews	Spanish	http://clic.ub.edu/corpus/es/node/106	18,000 customer reviews on hotels and restaurants from Hopinion
10	[38]	Multi-Domain	English	www2.cs.uic.edu/~liub/FBS/sentiment-analysis.html	6800 opinion words on 10 different products
11	TBOD [144]	Reviews	English		Product Review on Cars, Headphones, Hotels
12	[68]	Product Reviews	English	http://www.lsi.us.es/~fermin/index.php/Datasets	Product Reviews from Epinion.com on headphones 587 reviews, hotels 988 reviews and cars 972 reviews
13	[148]	Movie Reviews	Turkish	http://www.win.tue.nl/~mpechen/projects/smm/#Datasets	5331 positive and 5331 negative reviews on movie
14	[148]	Product Reviews	Turkish	http://www.win.tue.nl/~mpechen/projects/smm/#Datasets	700 +ve & 700 -ve reviews on books, DVD, electronics, kitchen appliances
15	ISEAR	English sentences	English	www.affective-sciences.org/system/files/page/2636/ISEAR.zip	The dataset contains 7666 such statements, which include 18,146 sentences, 449,060 running words.
16	[149]	Product Reviews	English	http://www.cs.jhu.edu/~mdredze/datasets/sentiment/	Amazon reviews on 4 domain (books, DVDs, electronics, kitchen appliances)
17	DUC data, NIST	Texts	English	http://www-nlpir.nist.gov/projects/duc/data.html , http://www.nist.gov/tac/data/index.html	Text summarization data
18	[70]	Restaurant and Hotel Reviews	English	http://uilab.kaist.ac.kr/research/WSDM11	Restaurant and Hotel Reviews from Amazon and Yelp
19	[114]	Restaurant Reviews	Cantonese	http://www.openrice.com	Reviews on restaurant
20	[125]	Biographical Articles	Dutch	http://www.iisg.nl/bwsa	574 Biographical articles
21	Spinn3r dataset	Multi-Domain	English	http://www.icwsm.org/2011/data.php	
22	[86]	Ironic Dataset	English	http://users.dsic.upv.es/grupos/nle/	3163 ironic reviews on five products
23	HASH [179]	Tweets	English	http://demeter.inf.ed.ac.uk	31,861 Pos tweets, 64,850 Neg tweets, 125,859 Neu tweets
24	EMOT [179]	Tweets and Emoticons	English	http://twittersentiment.appspot.com	230,811 Pos & 150,570 Neg tweets
25	ISIEVE [179]	Tweets	English	www.i-sieve.com	1520 Pos tweets, 200 Neg tweets, 2295 Neu tweets
26	[177]	Tweets	English	e-mail: apoorv@cs.columbia.edu	11,875 tweets
27	[52]	Opinions	English	http://patientopinion.org.uk	2000 patient opinions
28	[96]	Tweets	English	http://goo.gl/UQvdx	667 tweets
29	[39]	Movie Reviews	English	http://ai.stanford.edu/~amaas/data/sentiment/	50,000 movie reviews
30	[164]	Tweets	English	http://cs.stanford.edu/people/alecmgo/trainingandtestdata.zip	
31	[210]	Spam Reviews	English	http://myleott.com/op_spam	400 deceptive and 400 truthful reviews in positive and negative category. Last Accessed by: 12 April, 2015
32	[230]	Sarcasm and nasty reviews	English	https://nlds.soe.ucsc.edu/iac	1000 discussions, ~390,000 posts, and some ~73,000,000 words

More datasets

- SNAP review datasets: <http://snap.stanford.edu/data/>
- Yelp dataset: http://www.yelp.com/dataset_challenge/
- More on Twitter datasets, including critical appraisal: Saif et al. (2013)



Google dataset search

<https://toolbox.google.com/datasetsearch>



Q sentiment X

About



Feedback

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Sentiment Analysis Single Word

data.world

Updated May 22, 2018



Sentiment Analysis in Text

data.world

Updated Oct 24, 2019



Weather sentiment

data.world

Updated Oct 28, 2019



Economic sentiment indicator

data.europa.eu
db.nomics.world

Updated Nov 15, 2019



VnEmoLex: A Vietnamese emotion lexicon for sentiment...

zenodo.org
search.datacite.org

Published Jun 1, 2017

Emoji Sentiment Ranking

Explore at figshare.com

63 scholarly articles cite this dataset ([View in Google Scholar](#))

Unique identifier

<https://doi.org/10.6084/m9.figshare.1600931.v1>

Dataset created Nov 12, 2015

Dataset published Nov 12, 2015

Dataset updated Jan 20, 2016

Dataset provided by

figshare

Authors

Igor Mozetic; Petra Kralj Novak; Jasmina Smailović; Borut Sluban

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Description

A lexicon of 751 emoji characters with automatically assigned sentiment. The sentiment is computed from 70,000 tweets, labeled by 83 human annotators in 13 European languages. The Emoji Sentiment Ranking web page at http://kt.ijs.si/data/Emoji_sentiment_ranking/ is automatically generated from the data provided in this repository. The process and analysis of emoji sentiment ranking is described in the paper: P. Kralj Novak, J. Smailović, B. Sluban, I. Mozetič, Sentiment of Emojis, submitted; arXiv preprint, <http://arxiv.org/abs/1509.07761>, 2015.



V

9

Google dataset search

<https://toolbox.google.com/datasetsearch>

Google Dataset Search

opinion



100+ results found



The Opinions of the Committee of the Regions

data.europa.eu
data.wu.ac.at

Updated Jul 25, 2019



Opinion of the French on the relevance of their research onlin...

www.statista.com



European Data Protection Supervisor opinions

data.europa.eu

Updated Jan 25, 2018



London 2012 Opinion Research

data.gov.uk

find-data-beta-staging.cloudapps.digital

Consumer Opinion Surveys: Consumer Prices: Future Tendency of Inflation: European Commission and National Indicators for Spain
CSINFT02ESM460S

Explore at FRED

Dataset updated Nov 13, 2019

License

https://research.stlouisfed.org/fred_terms.html#copyright-citation-required

Description

Graph and download economic data for Consumer Opinion Surveys: Consumer Prices: Future Tendency of Inflation: European Commission and National Indicators for Spain (CSINFT02ESM460S) from Jun 1986 to Oct 2019 about consumer sentiment, consumer prices, Spain, consumer, and inflation.

Quality of the data may vary: labels and their interpretations, annotations may differ (crowd, students, linguists, derived from ratings). Be critical about the system and the data you use.

Transformer models

- <https://huggingface.co/models>

 **Hugging Face**

Models Datasets Pricing Resources Log In Sign Up

Tasks

Fill-Mask Question Answering Summarization Table Question Answering Text Classification Text Generation Text2Text Generation Token Classification Translation Zero-Shot Classification + 4

Libraries

PyTorch TensorFlow + 9

Datasets

wikipedia squad c4 bookcorpus dcep europarl jrc-acquis CLUECorpusSmall oscar squad_v2 + 193

Languages

en es fr sv fi de multilingual zh + 327

Licenses

apache-2.0 mit gpl-3.0 + 12

Models 89 sentiment

Sort: Most Downloads

cardiffnlp/twitter-roberta-base-sentiment
Text Classification • Updated Nov 13, 2020 • 55k

nlptown/bert-base-multilingual-uncased-sentiment
Text Classification • Updated Dec 11, 2020 • 51k

finiteautomata/beto-sentiment-analysis
Text Classification • Updated Oct 14, 2020 • 5,305

oliverguhr/german-sentiment-bert
Text Classification • Updated Dec 11, 2020 • 4,444

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m3hrdadfi/albert-fa-base-v2-sentiment-binary
Text Classification • Updated Dec 26, 2020 • 1,134

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Updated Jul 15, 2020 • 510

m3hrdadfi/albert-fa-base-v2-sentiment-deepsentip...
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Text Classification • Updated Dec 26, 2020 • 400

gilf/english-yelp-sentiment
Text Classification • Updated Aug 3, 2020 • 268

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Text Classification • Updated Jan 10 • 244

mrm8488/t5-base-finetuned-imdb-sentiment
Text2Text Generation • Updated Dec 11, 2020 • 173

seduerr/sentiment
Text2Text Generation • Updated Jan 12 • 168

wietsedv/bert-base-dutch-cased-finetuned-sentime...
Text Classification • Updated Dec 9, 2020 • 154

How to use transformer model for sentiment

- from transformers import **pipeline**
- sentimentenglish = pipeline("sentiment-analysis",
model="distilbert-base-uncased-finetuned-sst-2-english",
tokenizer="distilbert-base-uncased-finetuned-sst-2-english")
- sentimentenglish("Nice hotel and the service is great")
 - [{'label': 'POSITIVE', 'score': 0.9998814463615417}]
- sentimentenglish("The rooms are dirty and the wifi does not work")
 - [{'label': 'NEGATIVE', 'score': 0.9997869729995728}]

How to use transformer model for sentiment (Dutch)

- from transformers import **pipeline**
- sentimentdutch = pipeline("sentiment-analysis",
model="wietse/v/bert-base-dutch-cased-finetuned-sentiment",
tokenizer="wietse/v/bert-base-dutch-cased-finetuned-sentiment")
- sentimentdutch("Mooi hotel en de service is geweldig")
 - [{'label': 'pos', 'score': 0.9999955892562866}]
- sentimentdutch("De kamers zijn smerig en de wifi doet het niet")
 - [{'label': 'neg', 'score': 0.6675440073013306}]

Survey Literature

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